# QUÍMICA

Q1.- $5\overset{2+}{Fe}\overset{1-}{Cl_2} + \overset{1+}{K}\overset{7+}{Mn}\overset{2-}{O_4} + 8\overset{1+}{H}\overset{1-}{Cl} \longrightarrow 5\overset{3+}{Fe}\overset{1-}{Cl_3} + \overset{2+}{Mn}\overset{1-}{Cl_2} + \overset{1+}{K}\overset{1-}{Cl} + 4\overset{1+}{H_2}\overset{2-}{O}$

$\hookrightarrow$ OX $\quad \hookrightarrow$ RE.

AG. RE $\quad$ AG. OX.

$\overset{2+}{Fe} + Cl^{1-} + \overset{1+}{K} + \overset{7+}{Mn}O_4 \rightarrow \overset{3+}{Fe} + \overset{1-}{Cl_3} + \overset{2+}{Mn} + \overset{1-}{Cl_2}$

$5\cdot(\ \overset{2+}{Fe} = \overset{3+}{Fe} + 1\bar{e}\ )$

$1\cdot(\ \overset{7+}{Mn}O_4 + 8H^+ + 5\bar{e} = \overset{2+}{Mn} + 4H_2O\ )$

$5Fe^{2+} + \overset{7+}{Mn}O_4 + 8H^+ = 5Fe^{3+} + \overset{2+}{Mn} + 4H_2O$

K: 1 = 1
Fe: 5 = 5
Mn: 1 = 1
Cl: 18 = 18
H: 8 = 8
O: 4 = 4

$X = \dfrac{\text{Sust. oxidada} - \text{Sust reducida}}{\text{Agente reductor}} \Rightarrow X = \dfrac{5-1}{5} = \dfrac{4}{5}$ //

Resp. B)

Q2.- $KO_2 + H_2O + CO_2 \longrightarrow KH + HCO_3^- + O_2$

$100g\,\cancel{KO_2} \times \dfrac{1\,\cancel{mol\,KO_2}}{71\,g\,\cancel{KO_2}} \times \dfrac{1\,\cancel{mol\,CO_2}}{1\,mol\,\cancel{KO_2}} \times \dfrac{44\,\cancel{g\,CO_2}}{1\,\cancel{mol\,CO_2}} \times \dfrac{1\,\cancel{Kg\,CO_2}}{1000\,\cancel{g\,CO_2}} \times \dfrac{1\,\cancel{día}}{0{,}95\,\cancel{Kg\,CO_2}} \times \dfrac{24\,\cancel{h}}{1\,\cancel{día}} \times \dfrac{60\,min}{1\,h}$

$= 93{,}9\,min \approx 94\,min$

Resp. A) //

Q3.- $2KMnO_4 + 16HCl \rightarrow 2MnCl_2 + 2KCl + 5Cl_2 + 8H_2O$

$316{,}08\,g\,\cancel{KMnO_4} \times \dfrac{1\,mol\,\cancel{KMnO_4}}{158\,g\,\cancel{KMnO_4}} \times \dfrac{16\,\cancel{mol\,HCl}}{2\,mol\,\cancel{KMnO_4}} \times \dfrac{36{,}5\,g\,\cancel{HCl}}{1\,\cancel{mol\,HCl}} \times \dfrac{100\,g\,\cancel{HCl}}{36\,g\,HCl} \times \dfrac{1\,cm^3\,HCl}{1.2\,g\,\cancel{HCl}}$

$= 1352{,}19\,cm^3\,\cancel{HCl}$

Resp. B)

Q4.-

**DATOS**

$V_1 = 71\ ml$

$T_1 = 0°C \rightarrow 273,15\ K$

$P_1 = 1atm \rightarrow 760\ TORR$

$T_2 = 27°C \rightarrow 300.15\ K$

$P_T = 945,13\ TORR$

$P\dot{V}^{o}_{H_2O} = 26,7\ TORR$

$V_2 = ?\ ml$

$$P_T = P_2 + P\dot{V}^{o}_{H_2O}$$

$$P_2 = 945,13 - 26,7 = 918,43\ [TORR]$$

LEY COMBINADA DE LOS GASES.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1}{T_1} \cdot \frac{T_2}{P_2} = \frac{760\ \cancel{TORR} \cdot 71\ ml \cdot 300.15\ \cancel{K}}{273,15\ \cancel{K} \cdot 918,43\ \cancel{TORR}}$$

$$V_2 = 64,56\ [ml] \longrightarrow A$$

Q5.-

DATOS

$m_{BaCl_2 \cdot 2H_2O} = ?\,[g]$

$\bar{M}_{BaCl_2} = 208{,}33\ g/mol$

$\bar{M}_{BaCl_2 \cdot 2H_2O} = 244{,}35\ g/mol$

$m_{BaCl_2} = 50\ g$

$\% = 12$

$$50\,g\,BaCl_2 * \frac{12\,g\,BaCl_2}{100\,g\,BaCl_2} * \frac{1\,mol\,BaCl_2}{208{,}33\,g\,BaCl_2} * \frac{1\,mol\,BaCl_2 \cdot 2H_2O}{1\,mol\,BaCl_2} * \frac{244{,}35\,g\,BaCl_2 \cdot 2H_2O}{1\,mol\,BaCl_2 \cdot 2H_2O} =$$

$$m_{BaCl_2 \cdot 2H_2O} = 7{,}037\,[g] \rightarrow c$$